MRO Reverse Engineering Study

Assignment #8

Team 13

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Contents

1 Structure & Configuration Subsystems

The structural subsystem includes the support structure, mechanisms, launcher adapter, and moving parts. The structure and configuration of Mars Reconnaissance Orbiter (MRO) are driven by several factors, the most critical of which is the necessity of a dedicated nadir-facing instrument deck to fulfill the scientific objectives of the mission. This restriction in turn has driven the design of the structure and the various configurations of MRO throughout the mission. Additionally, the scientific instruments on the nadir deck have to be protected from intense aerodynamic stresses and thermal loads during aerobraking and plume impingement from thrusters during maneuvers, while avoiding direct solar exposure.

The primary structure is the main load-carrying structure, and thus is sized to withstand launch accelerations (-2 to 5g axially, 2g laterally [15]) and dynamic interactions. The primary structure of MRO is composed of deck plates aligned alongside central bus housing the propellant tank, as shown in Figure 1. These plates are referred to by their positions, namely: aft, middle, forward (HGA baseplate), and nadir decks. These decks are joined via support struts and gussets, integrated with the load-bearing central bus, which houses the propellant tank. The gussets stiffen the structure, i.e. raise the mode frequencies above the frequencies observed during launch and aerobraking, specifically, the low-frequency vibration that tends to be the design driver for spacecraft (S/C) structure, which constitutes the 0-50 Hz range [15]. The plates are made of aluminum honeycomb structures, and the central bus is made of titanium and reinforced with carbon-fiber reinforced polymer (CFRP) [10], which yields a structure high in mass-to-strength ratio.

The secondary structure of MRO includes deployables and supports for components, such as the solar arrays and SHAllow Subsurface RADar (SHARAD). The HIgh Resolution Imaging Science Experiment (HiRISE) is mounted via a rigid support ring connected to the middle deck, while the housing for the instrument itself is graphite [9].

The entirety of the S/C body and the High-Gain Antenna (HGA) are covered in Multi-Layer Insulation (MLI) blankets for thermal insulation, which provide little to no structural support, but add mass and thickness and affect the vibrational response of the S/C. The mass of the structure subsystem is calculated as the remainder when all of the previous subsystems are subtracted from the total mass, which is 352.25 kg, most of which is composed by the titanium tank.

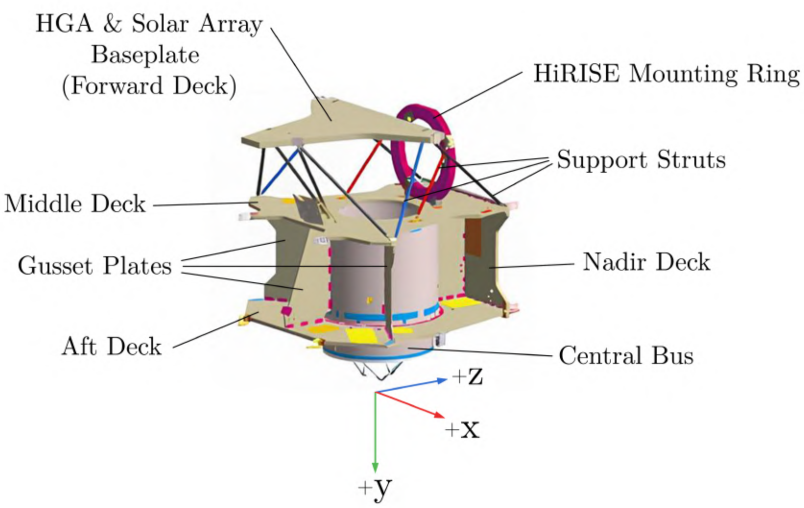

Figure 1: A CAD model of the structure of MRO, adapted from [10].

1.1 Mechanisms and Deployables

MRO employs several mechanisms to enable the deployment and pointing of its components. These include three two-axis gimbals: one for the HGA, providing precise and continuous Earth-pointing capability, and two for the solar arrays, enabling Sun-tracking throughout the mission. All of the gimbals are designed to withstand significant static and vibrational loads during launch, Mars Orbit Insertion (MOI) and aerobraking. Specifically, they shall resist and backdrive the reaction torques observed during all phases. For each axis, the gimbals utilize a two-phase brushless DC motor with a direct drive to the wave generator of a size-32 cup-type harmonic gear [2]. The gimbal actuators with these specifications were chosen to —in addition to bearing the launch loads —minimize jitter and achieve smooth rotor velocity [2], both of which are critical for the quality of scientific imaging. Since the gimbals are all single points of failure, the usage of heritage materials and processes were greatly prioritized [2].

In addition, SHARAD, a 10-m dipole antenna made up of two 5-m foldable tubes, is stowed until the transition phase (as shown in Figure 5) and utilized from Primary Science Phase (PSP) onward. SHARAD is made using elastic fibers, and thus self-deploys once released via solenoid-controlled actuators [14].

1.2 Configuration Across Phases

To protect the S/C during ascent and partially during LEOP, it is enclosed by a payload fairing (PLF). MRO is launched within the 4.2-m diameter PLF connected to the Centaur (the upper stage of the launch vehicle Atlas V 401 [15]), shown in Figure 2. The folded configuration is therefore driven by the inner dimensions of the PLF.

Figure 2: MRO in its launch configuration, next to the Centaur payload fairing [13].

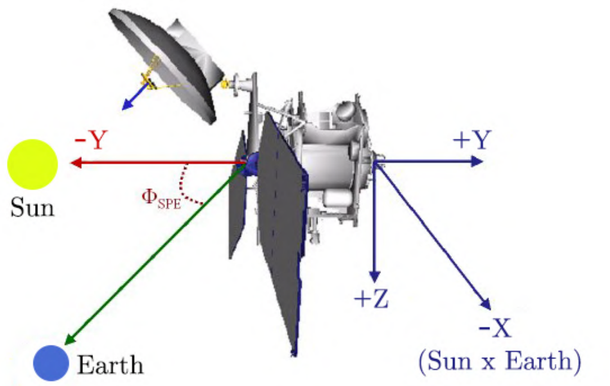

Figure 3: Cruise configuration of MRO, adapted from [16].

MRO is connected to the Centaur via an aluminum 1.2-m diameter payload adapter on the +Y face of the S/C. The payload adapter, positioned at the end point of the propellant tank, transmits the launch loads to MRO, and is jettisoned away along with the payload separation ring using a releasable clamp band after launch, separating with springs [12].

The solar arrays are in a folded position at launch (Figure 2), are deployed during checkout, and assume different configurations depending on mission phase. Specifically, they are in a spread-eagle configuration during aerobraking to maximize drag surface and during cruise (as shown in Figure 3, in *SHARAD Park position* (pointing away from SHARAD) during orbital night and occultations to

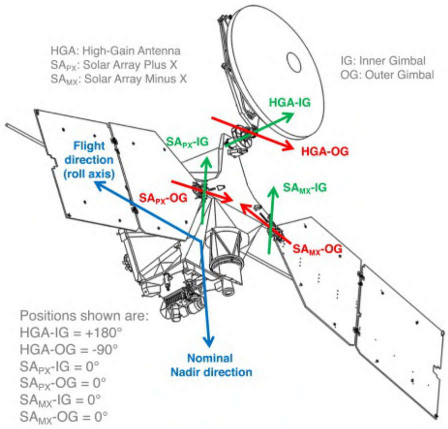

Figure 4: Nominal configuration of MRO during PSP and EM [3].

minimize electromagnetic interactions between the arrays and the instrument [3], and are nominally Sun-tracking during orbital days in PSP and Extended Mission (EM), as shown in Figure 4.

1.3 External Components and Distribution of Appendages

The solar arrays of MRO are mounted at the +X and -X edges of the forward deck. This results in a balanced mass distribution and symmetric aerodynamic loading, which help with the positioning of the Center of Mass (CoM) and stability during aerobraking respectively. During nominal operations, owing to both the 3pm Local Solar Time (LST) orbit and the lateral placement of the arrays, self-shadowing is minimal. The solar arrays remain fixed during the cruise phase as the AOCS maneuvers MRO to point the arrays at the Sun, with no self-shadowing of critical systems throughout this phase [16]. This configuration is therefore optimal for ensuring continuous solar exposure over the entire mission.

HGA is mounted on the HGA baseplate (forward deck), and is designed to stow flush against the -Z face of the body at launch, deploying perpendicularly away from the body during the aerobraking phase (in conjunction with the solar arrays) to maximize the drag surface while achieving a stable aerodynamic configuration. In this configuration, the center of pressure is aligned with the CoM along the Y-axis of the craft, and the aft deck nominally faces against the direction of flow.

SHARAD is placed at the +Y edge of the -Z face, which is permanently in shadow during nominal operations as a result of the 3pm LST orbit. In addition to minimizing thermal fluctuations and solar noise, this placement is furthest away from the solar arrays, which are the components that most impede the signal performance of the radar sounder [3].

The scientific instruments (shown in Figure 5), with the exception of SHARAD, are all situated on the nadir deck, pointing towards the +Z direction. The components on the nadir deck are not at risk of damage during the aerobraking, as they are downstream of the shock generated near the aft deck, where the energy content of the flow is lower [4].

The position of the Optical Navigation Camera (ONC), which is towards the +Y edge of the -Z face (next to SHARAD) is informed by the cruise attitude, as it shall point towards Mars during approach to measure the locations of its satellites and ascertain distance [8]. Another engineering instrument, the Ultra High Frequency (UHF) antenna is also situated on the nadir deck, which facilitates communication with surface vehicles on Mars. Placement of the UHF antenna in the vicinity of the instruments results in electromagnetic interference between these components, therefore the UHF antenna is activated when the science instruments are not and vice versa [5].

While there is no publicly available information regarding the distribution of the Thermal Control Subsystem (TCS), passive cooling radiators are presumed to be mounted on the aft deck, facing deep space throughout nominal operations.

Star trackers similarly function when facing deep space, and should be kept from direct solar exposure. As can be seen in technical drawings of MRO [16], the two redundant star trackers are positioned at the -Z face of the body (zenith direction), and nominally face deep space throughout scientific operations, relieving the spacecraft of the necessity to frequently execute attitude data acquisition maneuvers. Due to its orbit, the star trackers do not face the Sun and can function at any section of the flight.

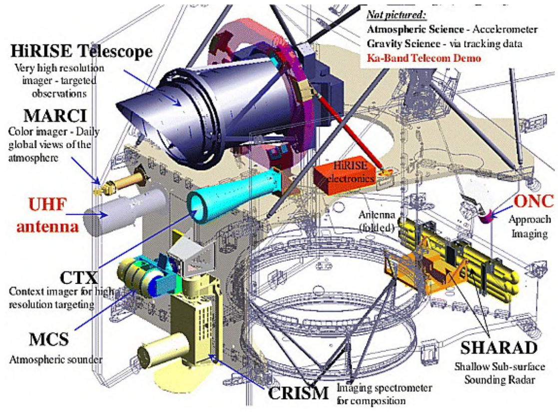

Figure 5: The science and engineering payloads onboard MRO at launch configuration. Middle and nadir deck plates are colored [17].

The six main thrusters are placed axisymmetrically around the centerline of the propellant tank [16], such that the only residual torque is generated is due to mounting error and thrust imbalances. Whereas the secondary thrusters are positioned in a coupled manner on the outer edges of aft and middle decks [16], resulting in a large lever arm and plume flow distant from the appendages.

1.4 Internal Components

The reaction wheel assembly (RWA), consisting of four wheels, (three orthogonal for full three-axis control, and one redundant [6]) is likely placed along the body frame axes of the S/C, near the CoM as such a positioning would yield a simpler control law and lower control torques. Wheels generate significant tonal and broadband vibrations due to mass imbalances and manufacturing imperfections, relating to the structural modes of the wheels. As many of the science instruments (specifically HiRISE) have tight jitter requirements [7], a detailed analysis of the wheel disturbances is required to ensure structural stability.

The placement of the propellant tank is constrained by a need to maintain a manageable CoM, as the CoM shifts with propellant consumption. As can be seen in Figure 1, it is axially aligned through the Y-axis of MRO. Using a simplified CAD model (introduced in Assignment 4), it was confirmed that due to the approximately in-line alignment of the propellant tank allows the net thrust vector of the main thrusters to be aligned with the CoM throughout the mission.

The On Board Data Handling (OBDH) components and electronic units of the instruments are housed primarily on the middle and aft decks [11]. Having the electronics situated in close proximity to each other minimizes line losses. The batteries are presumably mounted centrally, in a position which allows both reduced harness lengths (thereby lower voltage drops), and harness routing that minimizes inductive coupling with other components, while not increasing control torques drastically.

1.5 End-of-Life Constraints

Any end-of-life (EoL) considerations regarding planetary protection provisions have not affected the design of MRO, since all of its components, including the structure subsystem have been analyzed to exceed the temperature limit to be considered sterile upon Mars entry [1].

References

[1] J. Barengoltz and J. Witte. “Planetary protection implementation on Mars Reconnaissance Orbiter mission”. In: *Advances in Space Research* 42.6 (2008), pp. 1108–1119. ISSN: 0273-1177. DOI: https://doi.org/10.1016/j.asr.2007.10.001. URL: https://www.sciencedirect.com/science/article/pii/S0273117707010071.

[2] Edward A. Boesiger, ed. *38th Aerospace Mechanisms Symposium*. NASA Conference Publication NASA/CP-2006-214290. Accessed: 2025-05-23. Williamsburg, VA: NASA Langley Research Center, May 2006. URL: https://ntrs.nasa.gov/api/citations/20060028221/downloads/20060028221.pdf.

[3] Alessandro DiCarlofelice et al. “A new numerical model of the HF antenna of the Mars Reconnaissance Orbiter’s (MRO) Shallow Radar (SHARAD) and first results from a test at a high-angle spacecraft roll”. In: *Icarus* 419 (2024), p. 115802. ISSN: 0019-1035. DOI: https://doi.org/10.1016/j.icarus.2023.115802. URL: https://www.sciencedirect.com/science/article/pii/S0019103523003810.

[4] Michael Gallis et al. “Aerothermal Analysis for the Mars Reconnaissance Orbiter”. In: *37th AIAA Thermophysics Conference*. DOI: 10.2514/6.2004-2684. eprint: https://arc.aiaa.org/doi/pdf/10.2514/6.2004-2684. URL: https://arc.aiaa.org/doi/abs/10.2514/6.2004-2684.

[5] Dennis K. Lee Jim Taylor and Shervin Shambayati. *Mars Reconnaissance Orbiter Telecommunications*. Tech. rep. Article 12. NASA Technical Publication. Pasadena, California: Jet Propulsion Laboratory, California Institute of Technology, Sept. 2006. URL: https://descanso.jpl.nasa.gov/DPSummary/MRO_092106.pdf.

[6] M. D. Johnston et al. “The Mars Reconnaissance Orbiter Mission: From Launch to the Primary Science Orbit”. In: *2007 IEEE Aerospace Conference*. IEEEAC paper #1237, Version 4. Big Sky, Montana, USA: IEEE, 2007. DOI: 10.1109/AERO.2007.352742.

[7] S. W. Lee et al. “Mars Reconnaissance Orbiter Design Approach for High-Resolution Surface Imaging”. In: *Proceedings of the 26th Annual AAS Guidance and Control Conference*. February 5–9, 2003. American Astronautical Society, Rocky Mountain Section. Breckenridge, Colorado: AAS Publications Office, Feb. 2003.

[8] Andrew E. Lowman and John L. Stauder. “Stray light lessons learned from the Mars Reconnaissance Orbiter’s Optical Navigation Camera”. In: *Optical Systems Degradation, Contamination, and Stray Light: Effects, Measurements, and Control*. Ed. by John C. Fleming, Michael G. Dittman, and Philip T. C. Chen. Vol. 5526. International Society for Optics and Photonics. SPIE, 2004, pp. 240–248. DOI: 10.1117/12.566080. URL: https://doi.org/10.1117/12.566080.

[9] Alfred S. McEwen et al. “Mars Reconnaissance Orbiter’s High Resolution Imaging Science Experiment (HiRISE)”. In: *Journal of Geophysical Research: Planets* 112.E5 (2007). DOI: https://doi.org/10.1029/2005JE002605. eprint: https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1029/2005JE002605. URL: https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2005JE002605.

[10] NASA Jet Propulsion Laboratory. *Mars Reconnaissance Orbiter - Spacecraft Structures Subsystem*. Accessed: 2025-05-22, Archived: March 1, 2006. 2006. URL: https://web.archive.org/web/20060301180710/http://marsprogram.jpl.nasa.gov/mro/mission/sc_structures.html.

[11] NASA Jet Propulsion Laboratory. *PIA06756: Mars Reconnaissance Orbiter Taking Shape*. Accessed: 2025-05-23. 2005. URL: https://www.jpl.nasa.gov/images/pia06756-mars-reconnaissance-orbiter-taking-shape/.

[12] NASA Mars Exploration Program. *MRO Attached to Launch Vehicle*. Accessed: 2025-05-23. 2005. URL: `https://mars.nasa.gov/internal_resources/821/`.

[13] NASA/JPL-Caltech. *MRO Spacecraft Encapsulation*. Accessed: 2025-05-23. 2005. URL: `https://mars.nasa.gov/mro/images/KSC-MRO_Encapsulation-full.jpg`.

[14] Roberto Seu et al. "SHARAD sounding radar on the Mars Reconnaissance Orbiter". In: *Journal of Geophysical Research: Planets* 112.E5 (2007). DOI: `https://doi.org/10.1029/2006JE002745`.

[15] United Launch Alliance. *Atlas V Launch Services User's Guide*. Accessed: 2025-05-23. United Launch Alliance. Mar. 2010. URL: `https://www.ulalaunch.com/docs/default-source/rockets/atlasvusersguide2010.pdf`.

[16] Tung-Han You et al. "Mars Reconnaissance Orbiter interplanetary cruise navigation". In: *Proceedings of the 20th International Symposium on Space Flight Dynamics*. 2007.

[17] Richard W. Zurek and Suzanne E. Smrekar. "An overview of the Mars Reconnaissance Orbiter (MRO) science mission". In: *Journal of Geophysical Research Atmospheres* 112.E5 (May 2007). DOI: `10.1029/2006je002701`. URL: `https://doi.org/10.1029/2006je002701`.

Acronyms

AOCS Attitude and Orbit Control Subsystem. 3

CAD computer-aided design. 1, 4

CFRP carbon-fiber reinforced polymer. 1

CoM Center of Mass. 3, 4

DC Direct Current. 2

EM Extended Mission. 3

EoL end-of-life. 4

HGA High-Gain Antenna. 1–3

HiRISE HIgh Resolution Imaging Science Experiment. 1, 4

LEOP Launch and Early Orbit Phase. 2

LST Local Solar Time. 3

MLI Multi-Layer Insulation. 1

MOI Mars Orbit Insertion. 2

MRO Mars Reconnaissance Orbiter. 1–4

OBDH On Board Data Handling. 4

ONC Optical Navigation Camera. 3

PLF payload fairing. 2

PSP Primary Science Phase. 2, 3

RWA reaction wheel assembly. 4

S/C spacecraft. 1, 2, 4

SHARAD SHAllow Subsurface RADar. 1–3

TCS Thermal Control Subsystem. 3

UHF Ultra High Frequency. 3